303Modbus13 RTU Control

Python 3.10+
Modbus RTU
GitHub flexTech-KRK/303Modbus13
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Desktop GUI application for controlling and diagnosing the **Ideaflex 303Modbus13** relay module over **Modbus RTU** (serial/COM).

Git repository: github.com/flexTech-KRK/303Modbus13

> Designed for customers and integrators — quick module bring-up without a TCP/IP gateway.

Features

Feature	Description
2 relays	CH1–CH2 ON/OFF, bulk Both ON / Both OFF
2 opto inputs	Live status of IN1–IN2 (FC 0x02)
T/H sensor	Temperature and humidity readout (if installed)
Auto-refresh	Cyclic status polling every 1 s
Address change	Slave ID configuration via Modbus broadcast
Modbus console	Full access to FC 01–06, 0F, 10
Polish / English UI	Language selector in the header; choice saved in `settings.json`
Documentation	User guide · Register map

Requirements

- Windows 10/11 (or Linux with a serial port)
- Python 3.10 or newer
- USB ↔ RS-485 converter (or built-in COM port)
- 303Modbus13 module on the RS-485 bus

Quick start

1. Clone the repository

```
git clone https://github.com/flexTech-KRK/303Modbus13.git cd 303Modbus13
```

Online sources: README.md · REGISTERS.md

2. Virtual environment

```
python -m venv .venv # Windows .venv\Scripts\activate # Linux/macOS source .venv/bin/activate
```

3. Install dependencies

```
pip install -r requirements.txt
```

4. Run the application

```
python app.py
```

5. Connect to the module

1. Connect the module to a COM port (e.g. `COM4` via USB-RS485).
2. Select the COM port and click **Connect**.
3. Default Slave ID: `FF` (255, factory address).
4. Open the **Control** tab and test the relays.

Full UI walkthrough with screenshots: docs/USER_GUIDE.md

Communication parameters

Parameter	Value
Protocol	Modbus RTU
Baud rate	9600
Format	8N1
Slave ID (relays)	`0xFF` (255) factory default, then e.g. `0x01`
Slave ID (T/H sensor)	`0x02`
Timeout	**3.0 s** (serial port / pymodbus)

Retries	**3**
Post-write coil delay	0.35 s

At 9600 baud with a USB-RS485 converter, **do not use a 0.5–1.0 s timeout** — the module responds correctly but needs more time. Details, frames, and troubleshooting: **docs/REGISTERS.md**.

Opto inputs (IN / COM)

Inputs are **opto-isolated**. Both connections are required:

- **COM** → GND (0 V) of the signal source
- **INx** → signal voltage (+12…24 V DC relative to COM)

Input **COM** is **not** the same as the board power GND — it is a separate isolated circuit. Voltage on a relay contact alone, without wiring **INx** and **COM** on the module, will not produce `True` in Modbus (FC 0x02).

Command-line example (pymodbus)

Control relay CH1:

```
from pymodbus.client import ModbusSerialClient client = ModbusSerialClient( port="COM4",
baudrate=9600, parity="N", stopbits=1, bytesize=8, timeout=3.0, retries=3, ) client.connect()
# Turn CH1 ON (new module: slave=255 / 0xFF) client.write_coil(address=0, value=True,
slave=255) # Read opto inputs (count=8 in frame; result: IN1, IN2, …) inputs =
client.read_discrete_inputs(address=0, count=8, slave=255) print(inputs.bits[:2])
client.close()
```

Project structure

Path	Description
`app.py`	Tkinter GUI — application entry point
`i18n.py`	UI translations (Polish / English)
`locales/pl.json`, `locales/en.json`	Translation strings
`modbus_device.py`	Modbus RTU communication layer
`requirements.txt`	Python dependencies
`README.md`	This file
`scripts/generate_pdfs.py`	Build PDF documentation

`docs/USER_GUIDE.md`	User guide with screenshots
`docs/REGISTERS.md`	Full Modbus register map
`docs/images/`	Application screenshots
`docs/*.pdf`	PDF exports (README, USER_GUIDE, REGISTERS)

Generate PDFs

```
pip install markdown xhtml2pdf python scripts/generate_pdfs.py
```

Output: `docs/README.pdf`, `docs/USER_GUIDE.pdf`, and `docs/REGISTERS.pdf`.

First-time module setup

A new module ships with Slave ID **`0xFF` (255)**. To set address `0x01`:

1. Connect with Slave ID `FF`.
2. On the **Connection** tab, enter new address `01` and click **Save new address**.
3. Disconnect and reconnect with Slave ID `01`.

Alternatively — via the advanced console:

```
Function: Write Multiple Registers (0x10) Address: 0 Values: 1 Slave: 0
```

Modbus TCP/RTU gateway

The module can also be used through a network gateway (e.g. USR-M0) with Modbus TCP. The register map is identical — only the transport layer changes (IP instead of COM).

TCP/RTU application version: local repo `ModBus_Relay_303Modbus13` (RTU/COM version: [303Modbus13 on GitHub]).

Troubleshooting

Symptom	Solution
Cannot open COM port	Check Device Manager → COM ports; close other apps using the port
No response from device	Check RS-485 (A/B), Slave ID (`FF` or `01`), timeout **3 s**

Opto inputs always off	Wire **COM** → **GND** and **INx** → **+signal** (isolated circuit — see REGISTERS.md)
T/H sensor shows not connected	Sensor is optional — relay module works without it
Address change error	Use Slave=0 (broadcast)

License

Contact

If you have any problems or questions, contact **FlexTech_KRK** on Allegro or email **biuro@ideaflex.pl**.